

Robustness under Short Interaction Histories

Supplementary Analysis for Recommendation with Hidden Temporal Periodicity Modeling (RHYTHM)

Setting

For each held-out test target, we define the visible history length as $\ell_u = |\mathcal{S}_u^{\text{hist}}|$, i.e., the prefix actually available before predicting the test item rather than the user’s final full sequence length. Under leave-one-out evaluation, the visible prefix contains the training interactions plus the validation interaction, so users at the 5-core boundary have $\ell_u = 4$. We compare RHYTHM with two strong temporal baselines, TCPSRec and TALE, using NDCG@10 as the primary metric and HR@10 as an additional metric. Ta-Feng and MegaMarket use buckets 4–9, 10–19, 20–39, and 40+; DHRD uses 4–7, 8–11, 12–19, and 20+ because its interaction histories are substantially shorter. Relative gain is computed against the strongest baseline within each bucket.

Table 1: History-length distribution and density of phase-level evidence. Each cell reports “#users / non-empty phase ratio”.

Dataset	Short	Medium-short	Medium-long	Long
Ta-Feng	6,924 / 21.41%	7,092 / 29.37%	6,411 / 41.19%	5,734 / 61.93%
DHRD	21,544 / 9.86%	9,171 / 14.66%	7,395 / 19.69%	5,750 / 29.19%
MegaMarket	2,622 / 14.20%	2,435 / 20.34%	2,464 / 27.97%	5,725 / 45.99%

Results

Table 2: Performance by visible history length. Gain denotes the relative NDCG@10 improvement of RHYTHM over the strongest baseline in the same bucket.

Dataset	Length	NDCG@10				HR@10		
		RHYTHM	TCPSRec	TALE	Gain	RHYTHM	TCPSRec	TALE
Ta-Feng	4–9	0.1189	0.1065	0.1149	+3.50%	0.1802	0.1700	0.1648
	10–19	0.1172	0.1038	0.1129	+3.81%	0.1812	0.1723	0.1586
	20–39	0.1207	0.1077	0.1151	+4.94%	0.1891	0.1808	0.1669
	40+	0.1314	0.1125	0.1101	+16.79%	0.2032	0.1899	0.1608
DHRD	4–7	0.2721	0.2615	0.2358	+4.03%	0.4749	0.4660	0.3956
	8–11	0.2924	0.2693	0.2526	+8.57%	0.4889	0.4786	0.4213
	12–19	0.3011	0.2709	0.2617	+11.12%	0.5010	0.4847	0.4304
	20+	0.3220	0.2827	0.2761	+13.89%	0.5386	0.5104	0.4529
MegaMarket	4–9	0.1373	0.1325	0.0862	+3.65%	0.2018	0.1911	0.1354
	10–19	0.1322	0.1200	0.0913	+10.11%	0.2008	0.1918	0.1413
	20–39	0.1317	0.1152	0.1000	+14.36%	0.2050	0.1891	0.1469
	40+	0.1310	0.1092	0.0972	+19.88%	0.1991	0.1866	0.1485

Analysis

RHYTHM remains superior to both temporal baselines in every history-length group, including the shortest-history buckets. More importantly, its *relative* NDCG@10 advantage over the strongest baseline consistently expands as more history becomes visible: from 3.50% to 16.79% on Ta-Feng, from 4.03% to 13.89% on DHRD, and from 3.65% to 19.88% on MegaMarket. This pattern is particularly informative on DHRD, where the shortest group contains nearly half of the evaluated users and has only a 9.86% non-empty phase ratio, yet RHYTHM still improves over the strongest baseline.

The increasing relative gains coincide with a steadily increasing non-empty phase ratio (21.41%→61.93% on Ta-Feng, 9.86%→29.19% on DHRD, and 14.20%→45.99% on MegaMarket). This is consistent with RHYTHM’s mechanism: richer histories provide more behavioral evidence for discovering personalized temporal contexts and guiding historical aggregation. Since the buckets contain different user populations, we do not interpret absolute performance as a monotonic function of history length. Instead, the consistent result is that RHYTHM does not incur an additional disadvantage under short-history inference, while its advantage over strong temporal baselines becomes more pronounced when richer histories are available.